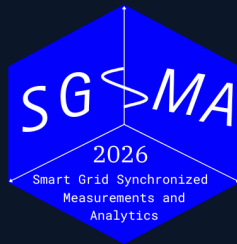


5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*



Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

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IBR disturbances expose a measurement gap

Event	Observed behavior	Monitoring limitation	Benchmark implication
Blue Cut Fire, CA 2016	Transmission faults caused nearly 1,200 MW of solar PV output loss; NERC linked part of the behavior to near-instantaneous frequency or voltage response.	The critical interval was a distorted, fault-clearing transient on a subsecond timescale.	Test whether estimators confuse waveform distortion with real frequency collapse.
Canyon 2 Fire, CA 2017	Two cleared faults produced 682 MW and 937 MW solar PV reductions.	NERC noted 4 s SCADA could not distinguish momentary cessation from tripping in some cases.	Report latency, settling, invalid samples, event-window behavior, and aggregate error.
Odessa events, TX 2021–2022	More than 1 GW-scale solar reductions followed normally cleared faults far from affected plants.	NERC found controls, protection, and model-fidelity gaps, including effects not captured well by positive-sequence models.	Stress tests must include control-like discontinuities, phase jumps, and model-mismatch regimes.
California BESS, 2022	Normally cleared faults caused abnormal BESS active-power reductions and inverter trips.	Causes included ac overcurrent/overvoltage, dc bus issues, unbalanced currents, and data-quality limits.	Evaluate frequency estimators under unbalance and non-sinusoidal contamination.

Source: NERC disturbance reports; NERC IBR disturbance monitoring white paper

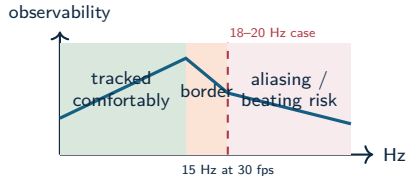
Standard PMU streams show consequences, not always causes

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i 2021 event exhibited 18–20 Hz oscillations after a generator trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.

Source: Kaua'i 18–20 Hz IBR oscillation analysis; NERC IBR monitoring white paper; IEC/IEEE 61850-118-1

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Operating claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- Estimator choice is conditional: event family, objective metric, and compute budget determine the preferred method.

Measurement rule

Report error, transient exposure, settling, latency, and cost as separate outputs for each estimator.

Technical contributions

Not claimed here

- PMU time-stamp certification;
- a single universal leaderboard;
- hardware latency measurements.

Contributions

- OpenFreqBench: a reproducible stress-test benchmark and platform for relay-class estimators;
- a metric-complete comparison protocol: RMSE, peak error, RoCoF error, T_{trip} , settling, and CPU;
- evidence that standard compliance tests miss IBR failure modes;
- an adapted RA-EKF as a tested estimator, with explicit $\dot{\omega}$ state and event gating.

The benchmark explains why estimator rankings change across objectives, regimes, and compute budgets.

Experiment scale

9+2
estimators

5
scenarios

4
families

grid
search

Central finding

No single estimator wins across objectives: the ranking changes with scenario, metric, and compute budget. Protection-class frequency measurement needs tests beyond compliance-style cases.

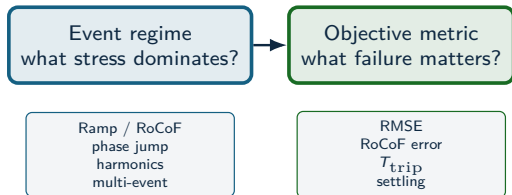
Dataset: 9 principal estimators, 2 legacy baselines, 5 stress scenarios, standard and IBR-specific metrics, plus an expanded sweep of 16 estimators, 34 scenarios, and 60 Monte Carlo runs per scenario.

Estimator selection logic

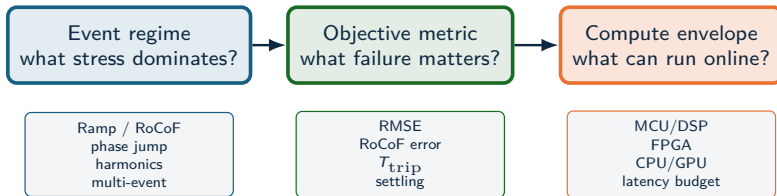
Event regime
what stress dominates?

Ramp / RoCoF
phase jump
harmonics
multi-event

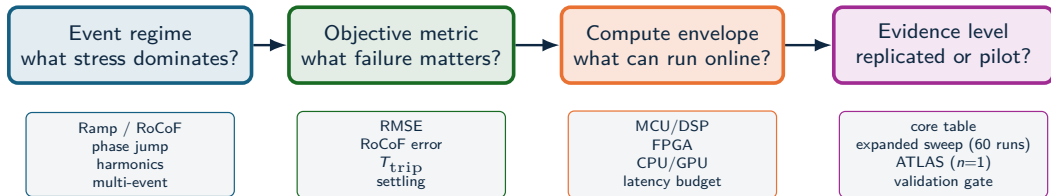
Estimator selection logic



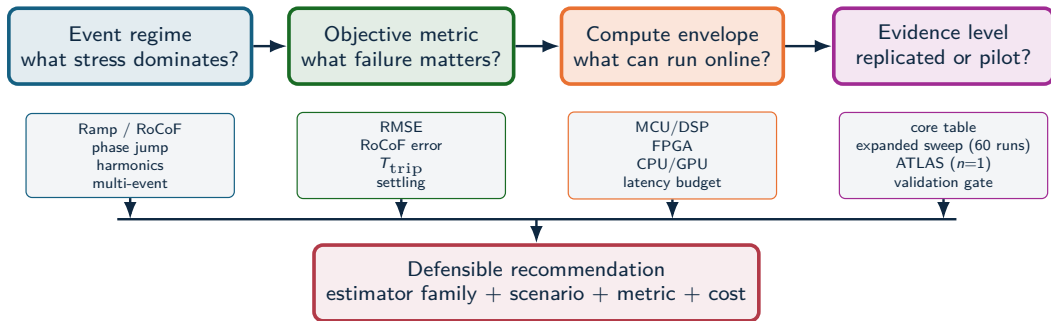
Estimator selection logic



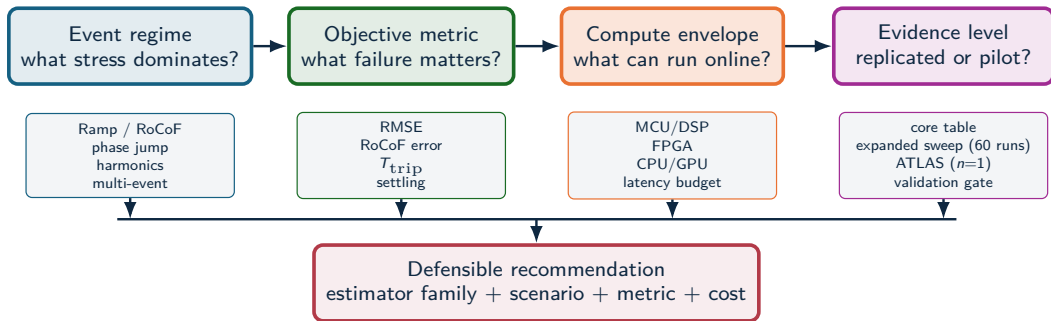
Estimator selection logic



Estimator selection logic



Estimator selection logic



Example: RA-EKF is recommended for ramp/phase-jump protection metrics under DSP/FPGA-class cost; global RMSE leadership remains scenario-dependent.

Benchmark literature gap

The field is mature in standards, metrology, open platforms, real datasets, and single-method studies. The missing layer is a reproducible, estimator-level benchmark that compares many algorithms under identical synthetic truth and deployment metrics.

Prior benchmark type	Strength	Gap for estimator-family comparison
IEC/IEEE PMU compliance	Defines synchrophasor, frequency, RoCoF, static/dynamic compliance tests	Leaves algorithm choice, family comparison, and CPU/latency outside the estimator report.
NIST PMU assessment/calibration	Provides metrological evaluation of PMU devices against standard requirements	Focuses on instruments and compliance behavior rather than a large matrix of open estimator implementations under controlled tuning policy.
OpenPMU / openPDC ecosystem	Enables open PMU construction and synchrophasor stream processing	Infrastructure-oriented, with no controlled estimator matrix using matched seeds and stress sweeps.
Open real PMU datasets	Supply realistic events for detection and application development	Truth is partially latent; hard to isolate causal stress factors or compute estimator error against known $f_{\text{true}}[k]$.
Single-study algorithm comparisons	Give detailed evidence for selected methods or scenarios	Usually limited estimator sets, limited artifact contract, and incomplete deployability metrics.

Compliance tests define requirements, operational data define realism, and controlled synthetic truth isolates estimator behavior under reproducible stress.

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

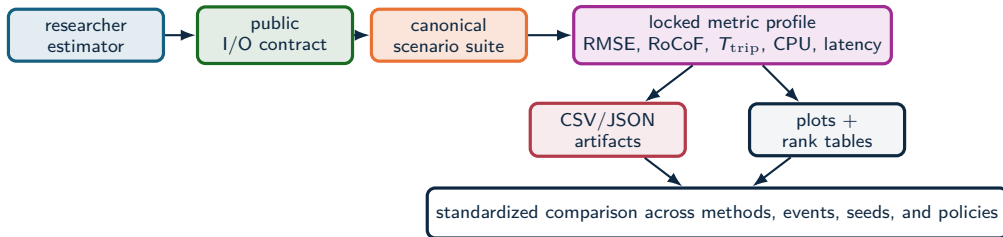
Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

Interpretive effect

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

OpenFreqBench platform



Researchers add

Estimator code, parameters, and hypotheses through the public contract.

Platform fixes

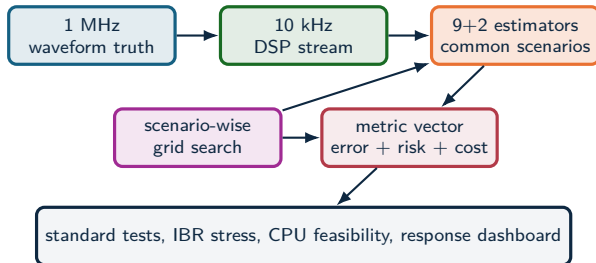
Truth signals, sampling, seeds, tuning policy, metric formulas, and schema.

Outputs

Benchmark reports, metric tables, plots, manifests, and readiness gates.

OpenFreqBench turns a new estimator into a repeatable benchmark entry instead of a one-off comparison.

Experimental method



Source: Methodology and test framework

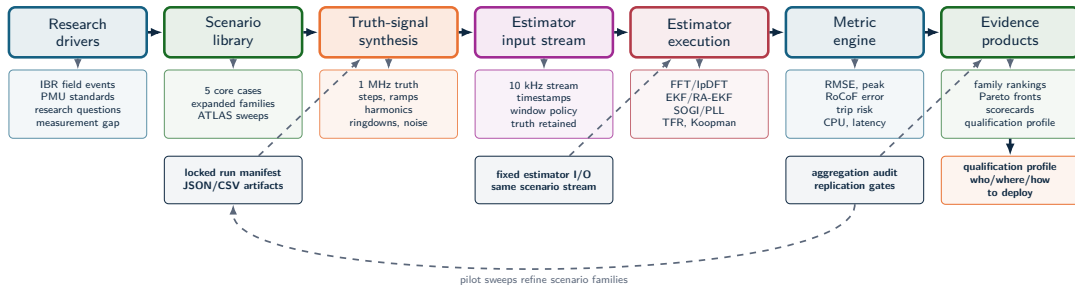
Measurement setting

Local, non-time-synchronized relay-class frequency tracking from a decimated 10 kHz stream; PMU timestamp certification is outside scope.

Experimental principle

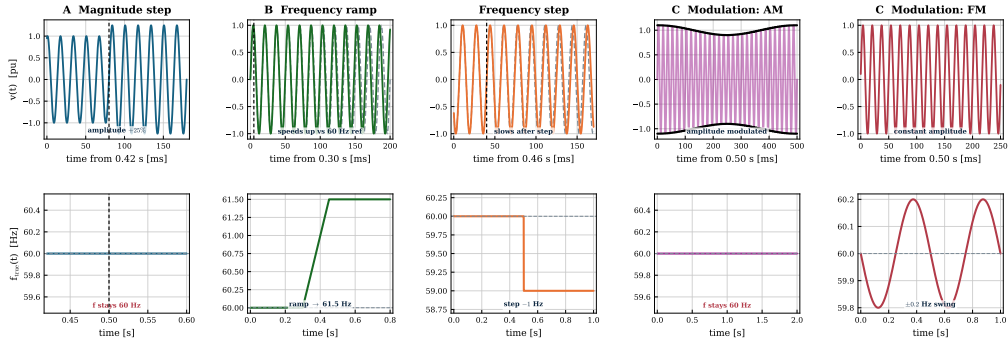
Each estimator–scenario pair is tuned within a declared grid, then compared using RMSE, peak error, trip exposure, settling, and CPU.

Full benchmark workflow



Scenario suite I — standard stresses

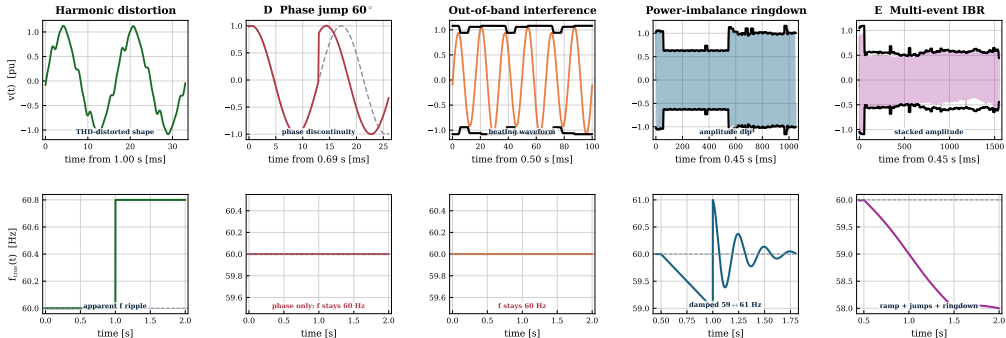
Scenario suite I — standard stresses: voltage $v(t)$ (top) vs. true frequency $f(t)$ (bottom)



Reading: each column pairs the voltage waveform $v(t)$ (top) with the true frequency $f(t)$ (bottom) for the same case. Magnitude step and AM leave $f(t)$ flat at 60 Hz — the stress is not a frequency event, yet estimators still err.

Scenario suite II — composite IBR stresses

Scenario suite II — composite IBR stresses: voltage $v(t)$ (top) vs. true frequency $f(t)$ (bottom)



Reading: harmonics, phase jumps and out-of-band interference also keep $f(t)$ near 60 Hz; only the ringdown and multi-event sequence move the true frequency. The waveform $v(t)$ is where most IBR stress actually lives.

Estimator families

Loop-based

SRF-PLL, SOGI-FLL.

Window-based

IpDFT, TFT.

Model-based / stochastic

EKF, UKF, RA-EKF.

Adaptive and data-driven

VFF-RLS, Koopman
(RK-DPMU), PI-GRU.

Legacy baselines

RLS and TKEO are retained as comparison references outside the recommended deployment set.

Interpretation rule

These families occupy different latency and stress-sensitivity positions. The comparison tracks where each family wins, degrades, and fails.

Metrics and decision variables

Error and recovery

$$e[k] = \hat{f}[k] - f_{\text{true}}[k]$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_k e[k]^2}$$

$$E_{\text{max}} = \max_k |e[k]|$$

Protection-oriented risk

$$\tau_{\text{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5 \text{ Hz}\}$$

$$t_s = \min\{t : |e[\tau]| \leq \epsilon, \forall \tau \geq t\}$$

$$C_{\text{CPU}} = \text{mean time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

RA-EKF mechanism

State augmentation

$$\begin{aligned}x_k &= [\theta_k, \omega_k, A_k, \dot{\omega}_k]^\top, & \omega &= 2\pi f \\ \theta_k &= \theta_{k-1} + \omega_{k-1}\Delta t + \frac{1}{2}\dot{\omega}_{k-1}\Delta t^2 \\ \omega_k &= \omega_{k-1} + \dot{\omega}_{k-1}\Delta t\end{aligned}$$

IBR-specific stress response

$$Q_k = \text{clip}(r_k, 0.25, 4) Q_{\text{base}}, \quad r_k = \frac{|\nu_k|}{\hat{\sigma}_\nu}$$

- **Covariance scaling:** inflate process noise under transients, deflate near steady state.
- **Event gating:** when $|\nu_k| > \gamma \hat{\sigma}_\nu$, softly reset θ_k while preserving ω_k and $\dot{\omega}_k$.

RA-EKF extends the EKF baseline with an explicit RoCoF state and event-aware covariance behavior for phase-jump recovery and steady-state tracking.

Standard scenario results

Method	Step +10%		Ramp +5 Hz/s		Modulation 2 Hz	
	RMSE	Peak	RMSE	Peak	RMSE	Peak
IpDFT	0.00157	0.00741	0.0729	0.157	0.000281	0.000747
TFT	0.00210	0.0190	0.0726	0.127	0.000260	0.000781
Koopman	0.000207	0.00523	0.0549	0.0860	0.000212	0.00178
SRF-PLL	0.000261	0.00132	0.142	0.354	0.000238	0.000441
SOGI-FLL	0.00900	0.0100	2.12	3.05	0.00900	0.0100
EKF	0.000733	0.00857	0.0180	0.0631	0.00119	0.00265
UKF	0.000774	0.00851	0.0207	0.0657	0.00116	0.00271
RA-EKF	0.00231	0.0299	0.0113	0.0441	0.00476	0.0125

Reading: the leader differs by scenario — Koopman and SRF-PLL lead the step and modulation cases, while RA-EKF leads the ramp (0.0113 Hz RMSE). No method dominates all three standard cases.

Source: Table III

Composite islanding: trip exposure under phase jump

Scenario D evidence

- Instantaneous $+60^\circ$ phase jump with harmonic and interharmonic content.
- Windowed leakage: IpDFT reaches 0.481 Hz RMSE; TFT reaches 2.74 Hz.
- EKF/UKF incur $T_{\text{trip}} = 0.165$ s without phase-reset mechanisms.
- SRF-PLL is competitive on RMSE, 0.0645 Hz, with $T_{\text{trip}} = 0$.

Benchmark reading: under this phase jump trip exposure spans 0–165 ms across methods; SRF-PLL and RA-EKF keep it near zero, while windowed and plain EKF/UKF methods do not.

RA-EKF result

$$\text{RMSE} = 0.0695 \text{ Hz}$$

$$T_{\text{trip}} = 0.6 \text{ ms}$$

$$T_{\text{trip}} : 165 \text{ ms} \longrightarrow 0.6 \text{ ms}$$

The gain comes from explicit $\dot{\omega}$ modeling plus event gating rather than post-hoc filtering.

Composite IBR sequence: why it is hardest

Scenario E stress components

- Harmonic-rich steady state: 5th at 5%, 7th at 3%, low interharmonics.
- $+40^\circ$ phase jump.
- -6 Hz/s RoCoF segment.
- $+80^\circ$ phase jump with ringdown.
- Sparse impulsive noise.

Standard-test gap

- The estimator sees a nonstationary waveform with stacked disturbances.
- Spectral leakage, loop re-lock, state-model mismatch, and recovery delay occur together.
- A method can be good on RMSE and still undesirable on trip exposure or CPU.

Once composite stress is shown to matter, ATLAS sweeps each stress component systematically instead of treating the multi-event case as a single aggregate disturbance.

Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μ s]
RA-EKF	0.511	0.141	54
EKF	0.789	0.118	18
Koopman	0.476	0.137	85
UKF	0.792	0.119	139
SRF-PLL	0.400	0.471	11
IpDFT	1.36	0.335	48
SOGI-FLL	0.568	0.422	25

Main reading

RA-EKF sits in the practical protection region: low trip exposure, moderate RMSE, and embedded-feasible CPU.

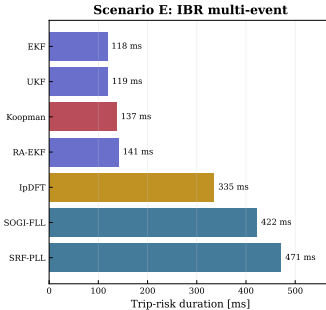
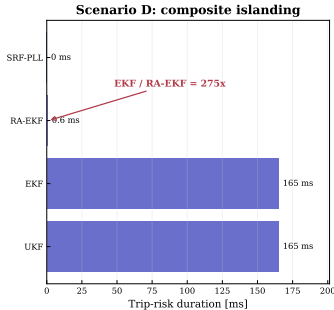
Source: Table IV

Scalar ranking failure

SRF-PLL has low CPU and good RMSE but much higher T_{trip} ; EKF has lower T_{trip} but worse RMSE; Koopman is accurate but costlier.

Trip-risk evidence: core stress cases

Trip-risk is an event metric, not an average-error metric

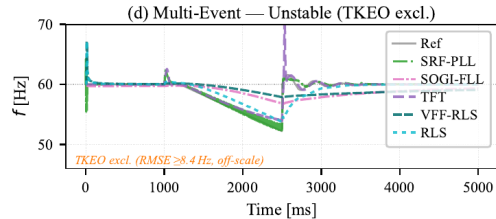
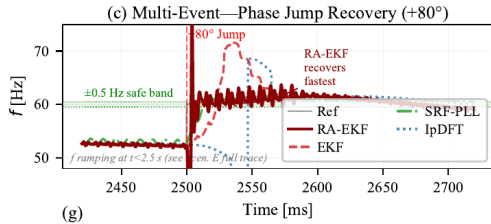
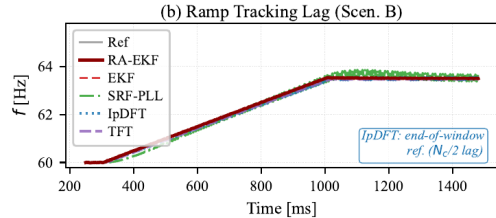
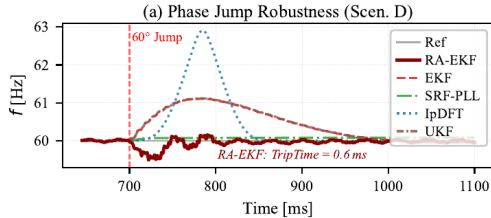


Result reading

- Scenario D exposes phase-reset behavior: RA-EKF reduces EKF exposure from 165 ms to 0.6 ms.
- Scenario E is multi-objective: RMSE and trip-risk select different methods.
- SRF-PLL is strong in Scenario D but reaches 471 ms in Scenario E.

Result: T_{trip} separates protection exposure from average error.

Event responses



Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	Koopman and SRF-PLL are strongest point cases	Step RMSE 0.000207 Hz; modulation peak 0.000441 Hz	Not enough for islanding or multi-event survival
Ramp and RoCoF tracking	RA-EKF is strongest in the standard ramp	Ramp RMSE 0.0113 Hz	Not a universal RMSE winner
Composite islanding exposure	RA-EKF removes long EKF exposure	0.6 ms vs 165 ms for EKF	SRF-PLL is also strong in this specific phase-jump case
Multi-event RMSE	Koopman and SRF-PLL are competitive	0.476 Hz and 0.400 Hz	RMSE hides trip-risk and compute trade-offs
Embedded protection feasibility	SRF-PLL, EKF, SOGI-FLL, RA-EKF remain plausible	11–54 μ s/sample range	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Computational feasibility

Method	Time/sample	Class	Target
SRF-PLL	11 μ s	P2	MCU
EKF	18 μ s	P1	DSP
TFT	25 μ s	P1	DSP
SOGI-FLL	25 μ s	P2	MCU/DSP
IpDFT	48 μ s	M1	DSP/FPGA
RA-EKF	54 μs	P1	DSP/FPGA
Koopman	85 μ s	M1	CPU
UKF	139 μ s	P1	high-perf CPU
PI-GRU	$\approx 136,000 \mu$ s	–	GPU only

Source: Table V

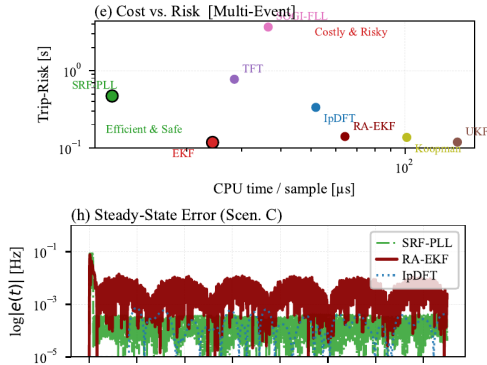
Deployment interpretation

The Python baseline separates feasible estimator families from clearly prohibitive ones; hardware latency requires platform timing.

Key comparison

RA-EKF is roughly three times EKF cost in Python but remains below the millisecond scale by a wide margin; PI-GRU is retained as a computational reference outside embedded protection targets.

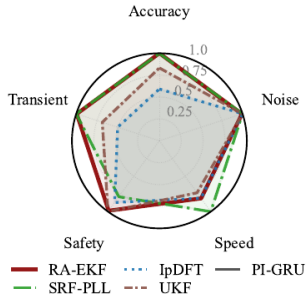
Deployment limits



Dashboard reading

- Deployment risk requires RMSE, trip exposure, settling, and delay.
- Trip-risk exposes the cost of estimator transients in composite events.
- Windowed methods can be accurate but delayed.
- RA-EKF occupies the main protection-oriented trade-off point.

Balance profile



Radar profile

- Accuracy, speed, noise immunity, safety, and transient behavior are conflicting axes.
- RA-EKF is selected for event-rich protection use cases.
- SRF-PLL remains valuable when cost and simplicity dominate.
- IpDFT is attractive where observation delay is acceptable.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Isl	Multi
PI-GRU	F	F	F	F	F
Koopman	P	F	P	F	F
TFT	P	F	P	F	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	F	P	P	F
SRF-PLL	P	F	P	F	F
EKF	P	P	P	F	F
UKF	P	P	P	F	F
RA-EKF	P	P	P	F*	F

Columns: **Isl** = composite islanding (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/ T_{trip} /settling threshold | **F*** borderline: meets trip-risk but misses the accuracy/settling target.

Several methods pass the standard-style Step/Ramp/Mod columns and still fail under composite islanding or multi-event IBR stress. Standard tests leave IBR event readiness partially untested.

Empirical claims

Protection behavior

An explicit dynamic state plus event gating cuts trip exposure.

Scenario D gives $T_{\text{trip}} = 0.6$ ms for the RA-EKF variant versus 0.165 s for the plain EKF.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved.

The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Metric disagreement

RMSE, trip-risk, and CPU select different winners.

Table IV separates Koopman/SRF-PLL accuracy, EKF/UKF trip exposure, and low-cost loop/filter deployment.

Deployability

Feasibility changes the accuracy ranking.

RA-EKF remains in a $54 \mu\text{s/sample}$ software-cost regime; PI-GRU is retained as a computational reference outside relay-class timing.

Expanded stress analysis

EXPANDED BENCHMARK: 16 estimators, 34 scenarios, 60 Monte Carlo runs per scenario

Core experiment

- 9 principal estimators + 2 legacy baselines.
- 5 scenarios: A–E.
- Standard, islanding, multi-event, and CPU results.
- RA-EKF supplies the tested estimator mechanism.

Expanded benchmark

- 16-estimator, 34-scenario, 10-family expanded sweep.
- 60 Monte Carlo runs per scenario.
- ATLAS severity curves and harmonic sweeps.
- Used to test sensitivity, scaling, and estimator-family stability.

The expanded benchmark scales the same measurement question across more event families, estimator types, and Monte Carlo replications.

Expanded benchmark: family winners

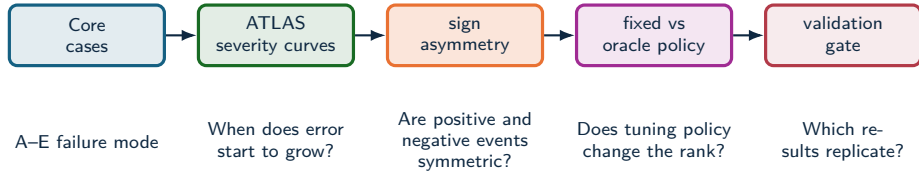
EXPANDED BENCHMARK: 16 estimators, 34 scenarios, 60 Monte Carlo runs per scenario

Family	Count	RMSE leader	RMSE	Trip-risk leader
IEEE ramps	9	SOGI-PLL	19.4 mHz	0 s, multiple estimators
Magnitude steps	7	EKF	0.388 mHz	0 s, multiple estimators
Modulation	3	RA-EKF	17.8 mHz	0 s, multiple estimators
Phase jumps	3	SOGI-FLL	174 mHz	ZCD, 0.0194 s
OOB interference	1	EKF	13.6 mHz	EKF / RA-EKF, 0 s
IBR harmonics	3	EKF	70.5 mHz	TFT / Prony, 0.00947 s
IBR ringdown	5	UKF	313 mHz	ESPRIT, 0.0989 s
IBR multi-event	1	PLL	173.5 mHz	IpDFT, 0.0439 s

The expanded sweep confirms the central pattern: estimator rankings depend on disturbance family and metric target.

ATLAS severity analysis

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: docs/ATLAS_PIPELINE.md; docs/ATLAS_METHOD_AUDIT.md

Reading the evidence: three levels

Diagnostic ($n < 30$)
mechanism + debugging only



Paper-grade ($n \geq 30$)
fixed policy, both signs



Journal-grade ($n \geq 100$)
cost reps, archived hashes

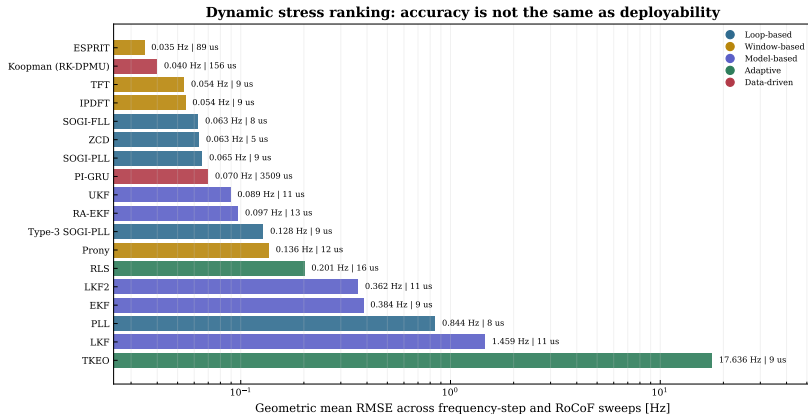
Where this talk sits

- Core 5-scenario results: tuned comparison evidence.
- Expanded sweep + ATLAS: **diagnostic pilots** ($n = 5$ / $n = 1$).
- They reveal *mechanisms* and rank *hypotheses* for the paper-grade campaign.

Every expanded/ATLAS result that follows is a pilot. The contribution is the benchmark and what it exposes — not a final leaderboard.

Dynamic accuracy ranking

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



When the textbook filter diverges

DIAGNOSTIC PILOT: small- n Monte Carlo, per-scenario tuned — paper-grade $n \geq 30$ run pending

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

Estimator	seed 1	seed 2	seed 3	seed 4	seed 5	trip-risk
EKF (textbook)	0.016	63.7	0.017	121.5	0.027	up to 1.35 s
RA-EKF	0.0055	0.0062	0.0083	0.0045	0.0058	0 s

What happens

The standard EKF tracks the ramp on most seeds but **intermittently diverges** to tens of Hz — the most dangerous failure mode for protection, because it is seed-dependent and silent.

Why RA-EKF matters here

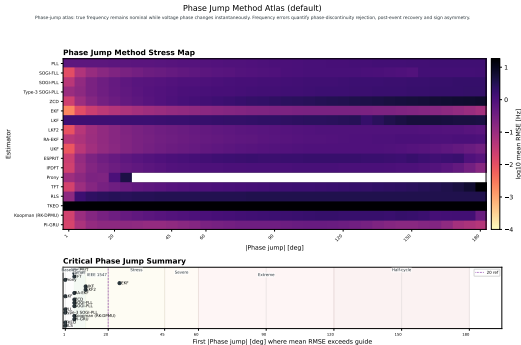
Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.

The standard UKF shows the same intermittent divergence on the 5 and 15 Hz/s ramps (up to 121 Hz on 2–3 of 5 seeds). The robustified state-space variant stays stable across the ramp family.

Source: [full_mc_benchmark/IEEE_Freq_Ramp/{EKF,RA-EKF}](#)

Phase-jump severity is non-monotone

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Worst case is not 180°

Most methods peak near 90–120° and *recover* toward 180° (a half-cycle jump partly self-cancels). EKF peaks at only 0.25 Hz @ 115°, back to 0.07 Hz @ 180°.

The TFT cliff

One exception: TFT rises gently, then jumps 2.5 → 21.9 Hz between 150° and 180° — a phase-reversal singularity.

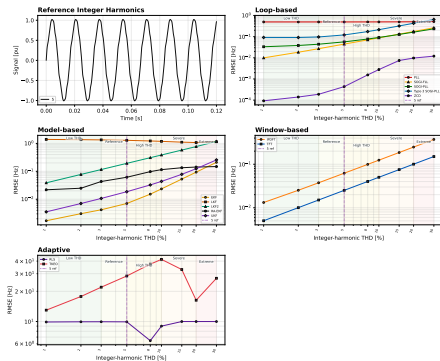
“Bigger jump = worse” is wrong for this family. Severity sweeps, not single test points, expose where each estimator actually breaks.

Harmonic degradation differs by estimator family

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

Integer Harmonics: RMSE [Hz] by estimator family (default)

Integer-harmonic isolation ranges: the true frequency is fixed at nominal and integer harmonic coefficients are isolated to a target THD. Interharmonics, subharmonics, impulses and frequency events are disabled so degradation can be attributed to harmonic distortion rather than mixed DSR artifacts.



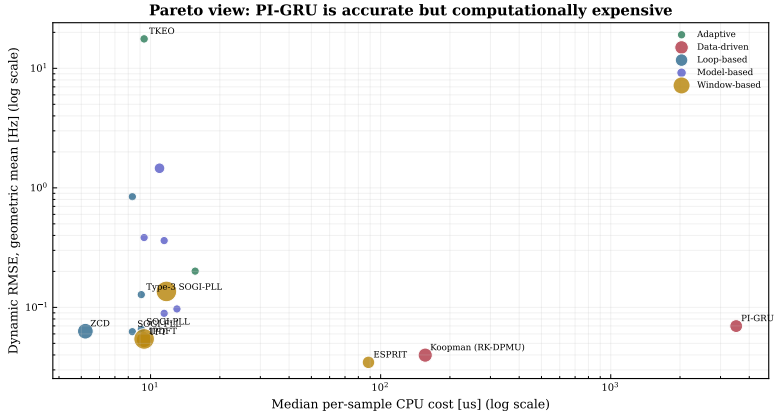
Grid-dashed lines mark perturbation sign where applicable. Bands show available run intervals/thresholds and regions are interpretation guides.

Reading

- Loop, window, model, and adaptive families degrade with THD at different rates.
- Best low-THD accuracy does not predict high-THD survival.
- Window methods hold up; some adaptive/model methods degrade steeply.

CPU–accuracy Pareto analysis

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Three findings

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2. **State-space estimation matters:** RA-EKF's explicit $\dot{\omega}$ state plus event gating reduces Scenario-D trip exposure from 0.165 s for EKF to about 0.6 ms.

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Estimator qualification includes transient recovery, trip exposure, compute feasibility, and standard-case RMSE.

Validity limits and next steps

Validation limits

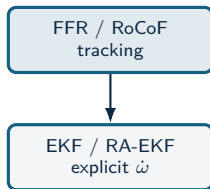
- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

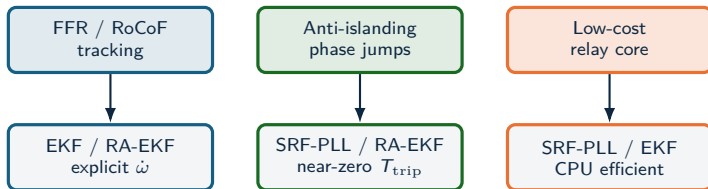
Recommendation map



Recommendation map



Recommendation map



Recommendation map

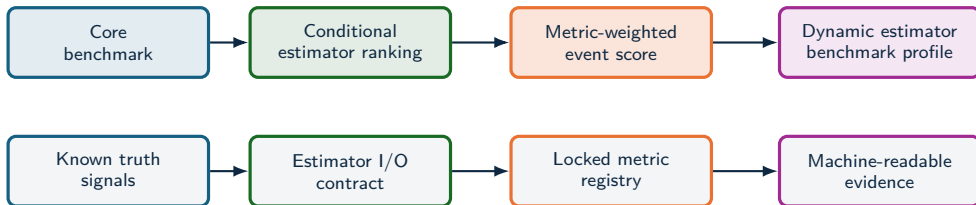


Recommendation map



Deployability depends on event class, trip-risk tolerance, and compute budget.

Qualification profile



Estimator selection moves toward **qualification** when each method is tested against a declared event profile, metric profile, and latency-cost envelope.

Standardization profile

Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.

IBR Event Qualification Score

COMPOSITE EVENT SCORE: weighted event-class qualification metric

Score for estimator e

$$\text{IBR-ERS}(e) = 1 - \sum_{s \in \mathcal{S}} w_s \Phi_s(e),$$

$$\Phi_s(e) = \alpha \tilde{E}_s + \beta \tilde{P}_s + \gamma \tilde{T}_s + \delta \tilde{S}_s + \eta \tilde{C}_s + \lambda \tilde{L}_s.$$

- $\tilde{E}_s$: normalized dynamic RMSE.
- $\tilde{P}_s$: normalized peak deviation.
- $\tilde{T}_s$: trip-risk duration.
- $\tilde{S}_s$: settling and recovery penalty.
- $\tilde{C}_s, \tilde{L}_s$: compute and latency penalties.

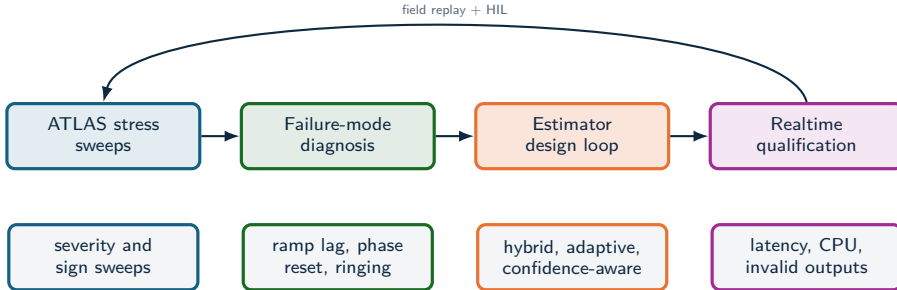
Composite metric

RMSE, trip-risk, transient recovery, and cost lead to different decisions. A composite score states the deployment objective explicitly.

Profile-specific weights

Protection-adjacent use can raise γ, δ, λ ;
forensic reconstruction can raise α, β ;
embedded monitoring can raise η .

Toward realtime IBR estimators



- The stress analysis identifies which physical stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and self-reported confidence.
- Validation should combine synthetic truth, EMT replay, hardware-in-the-loop, and field oscillography.

Validation agenda and conclusion

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

Result: estimator choice is tied to the event class, metric target, and compute envelope; the next validation step evaluates each method against declared event regimes, latency limits, and compute envelopes.

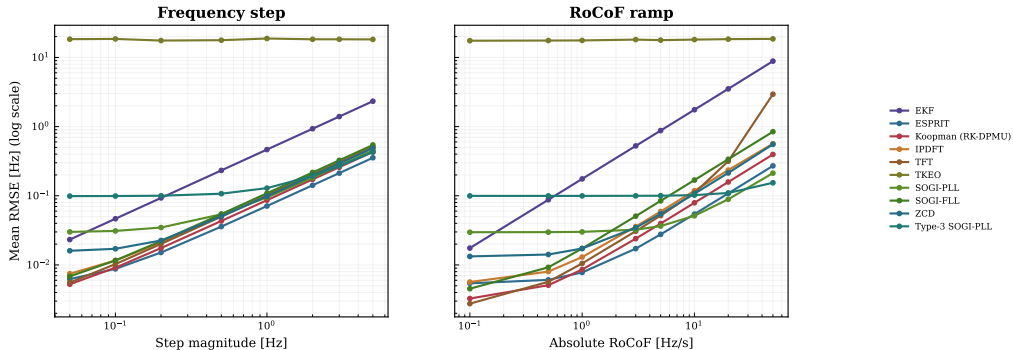
Backup

Metrics, extra stress maps, the
planned paper-grade run, references.

Sensitivity to disturbance severity

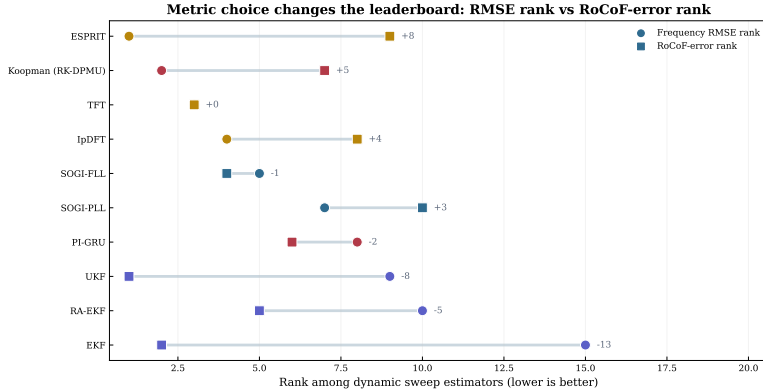
ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

Severity response curves reveal different failure modes



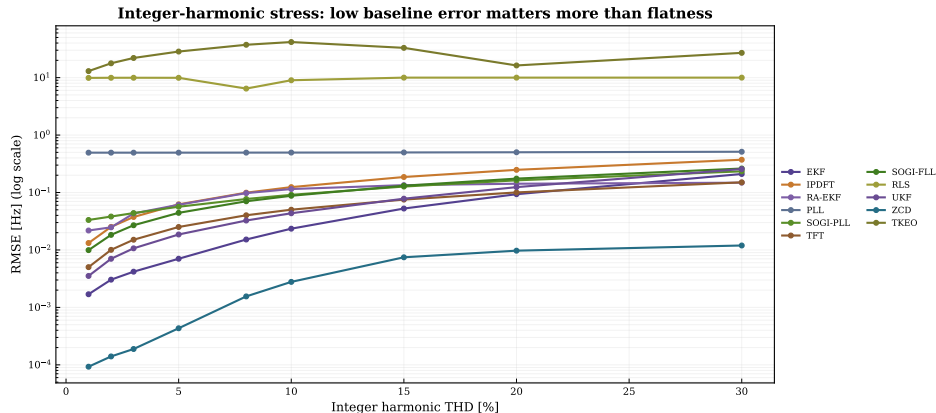
Rank shift when the metric changes

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Harmonic-stress response

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Backup: metric registry

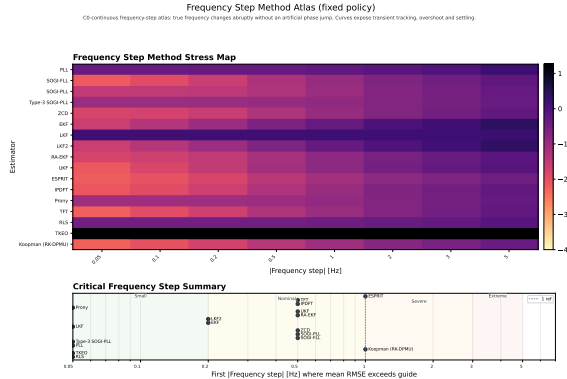
Metric	Definition
M1 RMSE	$\sqrt{\text{mean } e[k]^2}$ on steady-state-trimmed frequency error [Hz].
M3 peak	$\max_k e[k] $ [Hz].
M5 trip-risk	cumulative time with $ e[k] > 0.5$ Hz (protection-deadband risk proxy) [s].
M6 max-contig trip	longest <i>contiguous</i> deadband violation [s].
M8 settling	time until $ e $ stays ≤ 0.2 Hz [s].
M9 / M10 RFE	RoCoF error: robust max (99.5th pct) / RMS of $ \hat{f} - \dot{f} $ [Hz/s].
M13 CPU	execution time per sample [μ s] — software-cost proxy, not hardware latency.
M14 latency	declared structural delay (window / group delay) [ms].
M22 invalid rate	fraction of samples the estimator flags invalid / NaN.

Trip-risk is a protection-oriented risk proxy, not a relay trip command; CPU is a software-cost proxy. Differences below sampling/timing resolution are not interpreted.

Source: `src/analysis/metrics.py`; `docs/METHODS.md`

Backup: frequency-step stress map ($n = 30$, fixed policy)

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness still in progress



Reading

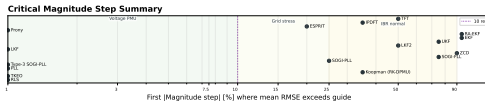
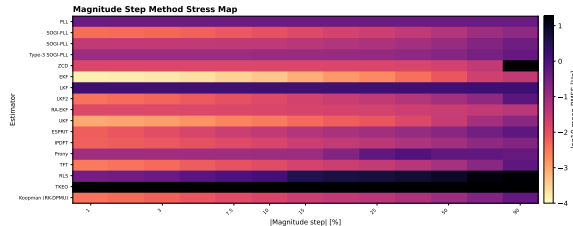
- **ESPRIT** leads almost every step; Koopman wins only the tiny ± 0.05 Hz steps.
- RMSE grows $\sim$ linearly with step size (≈ 0.07 Hz per 1 Hz).
- **TKEO** flat-broken (≈ 15 – 22 Hz).
- **Sign-symmetric**: 16/17 within $1.5\times$ — unlike magnitude steps and RoCoF.

Backup: magnitude-step stress map ($n = 30$, fixed policy)

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness still in progress

Magnitude Step Method Atlas (fixed policy)

Pure magnitude step: true frequency remains nominal. Frequency errors quantify AM-to-FM cross-sensitivity and numerical robustness. Upward steps are swells; downward steps are sags. The default sag grid steps below 100% to avoid a zero-voltage test.

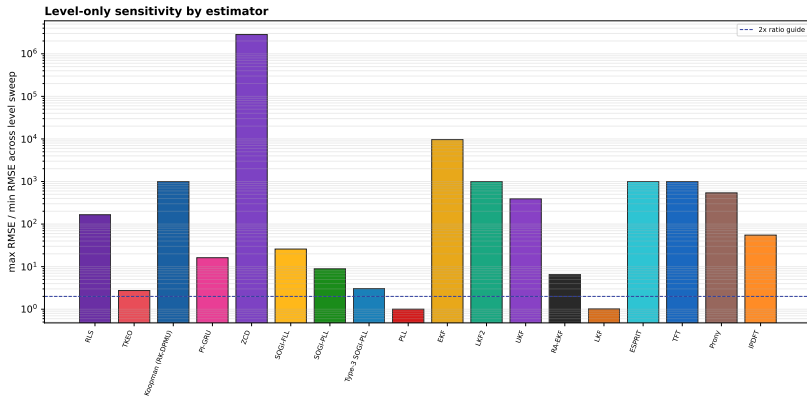


Reading

- EKF leads every step size (0.17–7.8 mHz).
- TKEO is flat-broken (≈ 18 Hz at all levels).
- ZCD fails on deep sags: 374 Hz at -90% vs 0.009 Hz at $+90\%$.
- Voltage sags are $\sim 2\times$ harder than swells of equal size.

Backup: noise robustness ranking

DIAGNOSTIC PILOT: small- n Monte Carlo, per-scenario tuned — paper-grade $n \geq 30$ run pending



Backup: the paper-grade run (planned)

Target command

```
python -m pipelines.atlas_sweep -sweeps all -policy
fixed_policy -n-runs 100 -n-cost-reps 3 -tune-trials 80
-output-subdir atlas-full-v1
```

- Full 18-estimator canonical set, fixed policy.
- Both signs, $n \geq 100$ Monte Carlo.
- Archived seeds, manifests, and hashes.
- Regenerate every table and figure from artifacts.

This converts today's diagnostic pilots into paper- and journal-grade claims through the readiness gate.

Source: docs/ATLAS_PIPELINE.md; docs/ATLAS_ROADMAP.md

Backup: selected references

Standards & platforms

- IEEE/IEC 60255-118-1:2018 — synchrophasor, frequency, RoCoF compliance.
- NISTIR 8106 — PMU performance assessment.
- OpenPMU, openPDC, FNET/GridEye, LBNL micro-PMU, PNNL PMU event library.

IBR field events

- NERC Blue Cut Fire (2016), Canyon 2 (2017), Odessa (2021–22), CA BESS (2022).
- NERC IBR disturbance-monitoring white paper.
- Kaua'i 2021 18–20 Hz IBR oscillation analysis.

Field cases define the stress modes; no claim is made that a single estimator would have prevented a disturbance.